

Business Analysis Research

Fundamentals of business analysis and a practical case study: subtitle options on YouTube

What Is Business Analysis?

Business analysis is the process of understanding the needs of an organization or project, defining problems, and developing solutions that create value for those problems.

What Are the Duties and Responsibilities of a Business Analyst?

A business analyst is the bridge between the business units and the technical team.

- Uncovering and analyzing the real needs of the business units.
- Translating these needs into detailed requirements that the technical team can understand.
- Modeling processes and identifying improvement opportunities.
- Making sure the developed solution meets the defined business goals.

Which Steps Does a Business Analyst Follow When Joining a Project?

- 1 **Current State Analysis:** Understanding the present structure of the project or company.
- 2 **Stakeholder Analysis:** Determining who needs to be communicated with.
- 3 **Requirements Elicitation:** Gathering needs through various methods.
- 4 **Analysis and Documentation:** Turning the gathered information into User Stories, Use Cases, or process flow diagrams.
- 5 **Approval and Development Support:** Obtaining stakeholder approval and guiding the software team throughout development.

What Is a Requirement and What Are Its Types?

A requirement is a feature, capability, or condition that a product or system must have in order to fulfill a specific purpose.

- **Business Requirements:** Why the company is undertaking this project (e.g., increasing sales by 20%).
- **User Requirements:** What users want to be able to do in the system.
- **System Requirements:** How the system will accomplish these (the functions of the software).

The Difference Between Functional and Non-Functional Requirements

- **Functional Requirement:** Defines what the system must do. (e.g., "The user must be able to add products to the cart.")
- **Non-Functional Requirement:** Defines how the system must do it. (e.g., "Adding a product to the cart must complete in under 1 second.")

Requirements Elicitation Techniques

- **Interview:** Gathering information from stakeholders through in-depth, question-and-answer sessions.
- **Workshop:** Bringing all relevant parties into one room to reach shared decisions and define flows.
- **Observation:** Watching users while they perform their own work.
- **Questionnaire/Survey:** Collecting quantitative data from a large number of users.
- **Document Analysis:** Reading existing procedures and reports.

As-Is and To-Be Processes

- **As-Is (Current State):** The present, untouched state of the system or way of working.
- **To-Be (Target State):** The design of how the process will work after the new project or software goes live.

Why Is Process Analysis Important?

Process analysis reveals bottlenecks, unnecessarily repeated steps, and manual work. The only way to define a problem correctly before producing a solution is to map the process as a whole.

User Story and Acceptance Criteria

- **User Story:** A short, non-technical requirement statement that focuses on the value perceived by the user.

User Story format

As a [User Role], I want to [Action I Want to Perform], so that [Benefit/Value I Will Gain].

- **Acceptance Criteria:** The test conditions a User Story must pass in order to be considered "Done." They define the boundaries of the work.

- **The difference:** A User Story describes the purpose and scope of the work, while Acceptance Criteria define the rules we will use to test whether the work was done correctly.

What Is a Use Case and When Is It Used?

A Use Case is a step-by-step interaction scenario between a system and the person using it. It is used especially in complex systems, when the main flow (Happy Path) and the alternative/error cases need to be shown in detail.

The Business Analyst's Communication Strategy

- **With the customer/business unit:** Speak entirely in terms of business processes and the value to be created, without software jargon.
- **With the developer:** Speak through clear business rules, data constraints, and logical flows, while respecting technical limits.
- **With the tester (QA):** Convey, through acceptance criteria, clear rules that form the basis for test scenarios.

Requirement Evaluation, Change, and Bug Analysis

- **Correct and complete requirement:** A requirement is successful if it follows the "INVEST" principle (Independent, Negotiable, Valuable, Estimable, Small, Testable) and all edge cases have been considered.
- **Requirement change:** The analyst does not immediately say "Yes." They perform an Impact Analysis. The cost of the change, its effect on the schedule, and conflicts with other modules are evaluated and presented for stakeholder approval.
- **Bug vs. Improvement:**
 - **Bug:** If the system behaves contrary to a previously written requirements document or acceptance criteria, it is a defect.
 - **Improvement:** The system works as designed and without errors, but an addition is made to improve the user experience or to meet a new business need; this is an enhancement.

What to Pay Attention to When Reviewing an Application from the User's Perspective

When exploring an application, a business analyst does not just press buttons; they question the logic of the process. They should ask questions such as: "Why did the user come here?", "Is there a step in the flow that blocks them?", "Are the error messages helpful and guiding?", and "How does the system react if I enter unexpected data?"

Practical Part: Subtitle Option on YouTube

User Scenario

The user wants to watch foreign-language content with subtitles in their own language.

As-Is Situation

The user wants to watch a video in a foreign language (e.g., English) on YouTube. The application can generate automatic subtitles for the language spoken in the video (e.g., English to English subtitles). However, when the user wants to watch this content with subtitles in their native language (e.g., Turkish), this option is only available if the content owner has manually added subtitles in that language. Since most videos have no such translated subtitles, the user either watches the video without any subtitles or settles for the automatic subtitles in the original language, which is useless for a user who does not know that language.

Problem

The language the user wants to understand does not match the subtitle language the system offers; the content remains inaccessible due to the language barrier.

To-Be Process Proposal

A language translation layer is integrated into the existing system, which automatically detects the original language of the video and generates automatic subtitles from speech. As a result:

- 1 The system first generates automatic subtitles in the original language.
- 2 The user can choose the desired target language from the subtitle settings.
- 3 The system automatically translates the generated original subtitles into the selected language and displays them in sync with the video.
- 4 This translation is offered by default on every video, without depending on the content owner adding manual subtitles.

User Story 1

User Story

As a YouTube user who does not know a foreign language, **I want to** see the automatic subtitles of the video I am watching translated into my own language, **so that** I can understand the content even if I do not know the language.

Acceptance Criteria

- **Given** automatic subtitles in the original language are available on a video, **When** the user selects their own language (e.g., Turkish) as the subtitle language, **Then** the subtitles must be translated into the selected language and displayed in sync with the video
- **Given** the user has turned on the translated subtitles, **When** the video continues to play, **Then** the translated subtitles must stay in sync with the speech with a delay of at most 1-2 seconds
- **Given** manual subtitles in that language have already been added by the content owner, **When** the user selects the same language, **Then** the system must show the manual/original subtitles with priority rather than the automatic translation (quality priority)

User Story 2

User Story

As a YouTube user, **I want to** be able to change the subtitle language easily while the video is playing, **so that** I can switch without interrupting the flow when I need to change to a different language during viewing.

Acceptance Criteria

- **Given** the video is playing, **When** the user selects a different language from the subtitle/settings menu, **Then** the subtitles must start appearing in the new language within 2 seconds at most, without the video pausing
- **Given** the user has selected a language, **When** they start watching another video from the same channel, **Then** the last selected subtitle language must be remembered as the default (preference persistence)
- **Given** the translated subtitles in the selected language could not be generated for a technical reason, **When** the user tries to select that language, **Then** a clear notice such as "Subtitles cannot be generated for this language right now" must be shown to the user (there must be no silent failure)

Bug Example

The user selects "Turkish" as the subtitle language, and the subtitles start appearing correctly in Turkish. However, when the user pauses the video, leaves the application, and later returns to the same video, the subtitle language is automatically reset to the video's original language - the user's selection is not preserved.

Improvement/Enhancement Example

Adding a "Dual Subtitles" feature to the application is an enhancement proposal. For example, the user can follow the original English on the top line and the automatically translated Turkish on the bottom line at the same time.

Questions the Business Analyst Conducting This Review Would Ask

- 1 Which translation engine (e.g., the Google Translate infrastructure) will be used for the automatic translation, and how will its accuracy rate be measured?
- 2 If manual subtitles and automatically translated subtitles both exist in the same language, which one will be prioritized, and by what criterion will this decision be made?
- 3 How will this translation feature work in real time during live streams, and what should the latency tolerance be?
- 4 Will the user's language preference be stored per account or per device?
- 5 Will content owners be held responsible for the quality/accuracy of the automatically translated subtitles, or will this be entirely the platform's responsibility?